

Risk register

Operational environment:

The bank is located in a coastal area with low crime rates. Many people and systems handle the bank's data—100 on-premise employees and 20 remote employees. The customer base of the bank includes 2,000 individual accounts and 200 commercial accounts. The bank's services are marketed by a professional sports team and ten local businesses in the community. There are strict financial regulations that require the bank to secure their data and funds, like having enough cash available each day to meet Federal Reserve requirements.

Asset	Risk(s)	Description	Likelihood	Severity	Priority
Funds	Business email compromise	<i>An employee is tricked into sharing confidential information.</i>	3	3	9
	Compromised user database	<i>Customer data is poorly encrypted.</i>	2	3	6
	Financial records leak	<i>A database server of backed up data is publicly accessible.</i>	3	3	9
	Theft	<i>The bank's safe is left unlocked.</i>	1	3	3
	Supply chain disruption	<i>Delivery delays due to natural disasters.</i>	1	2	2
Notes	<i>How are security events possible considering the risks the asset faces in its operating environment?</i> Security events are possible because the bank handles sensitive financial data and funds across both on-premises and remote work environments. Since many employees and systems have access to bank data, risks such as phishing, weak encryption, public database exposure, and theft could affect confidentiality, integrity, and availability. The bank's coastal location also creates a possible risk of supply chain disruption from natural disasters. <u>Why these scores make sense:</u> 1. Business email compromise — 9				

	This gets a high score because employees could be tricked into sharing confidential information, and the bank has many employees handling sensitive data. 2. Compromised user database — 6 Poor encryption is serious because customer data could be exposed. I gave it a moderate likelihood but high severity. 3. Financial records leak — 9 A publicly accessible backup database is very risky. Since the data is exposed and financial records are highly sensitive, both likelihood and severity are high. 4. Theft — 3 The safe being unlocked is serious, but the scenario says the bank is in a low-crime area, so the likelihood is lower. The severity is still high because funds could be stolen. 5. Supply chain disruption — 2 The bank is in a coastal area, so natural disasters are possible, but they are less frequent. The impact is moderate because delays could affect daily cash availability and operations.
--	---

Asset: The asset at risk of being harmed, damaged, or stolen.

Risk(s): A potential risk to the organization's information systems and data.

Description: A vulnerability that might lead to a security incident.

Likelihood: Score from 1-3 of the chances of a vulnerability being exploited. A 1 means there's a low likelihood, a 2 means there's a moderate likelihood, and a 3 means there's a high likelihood.

Severity: Score from 1-3 of the potential damage the threat would cause to the business. A 1 means a low severity impact, a 2 is a moderate severity impact, and a 3 is a high severity impact.

Priority: How quickly a risk should be addressed to avoid the potential incident. Use the following formula to calculate the overall score: **Likelihood x Impact Severity = Risk**

Sample risk matrix

		Severity		
Likelihood		Low 1	Moderate 2	Catastrophic 3
	Certain 3	3	6	9
	Likely 2	2	4	6
	Rare 1	1	2	3